

Examples and complements

SGA 1, Exposé XI

English draft

A. Grothendieck

About this draft

This is an unofficial English translation of Exposé XI, “Exemples et compléments”, from SGA 1. The source volume is A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental* (SGA 1), Séminaire de géométrie algébrique du Bois Marie, 1960–61. It follows the slightly corrected SMF recomposition, arXiv:math/0206203v2.

This draft contains the whole exposé, including proofs, footnotes, diagrams, the updating remarks added in 2003 by M. Raynaud (between bold brackets and marked (MR)) where the exposé has them, and its bibliography. Statement numbering and mathematical notation follow the source. References to other exposés print the source’s numbers. Apparent misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

The translator’s contribution is licensed under CC BY-SA 4.0. The French original remains copyright of its authors and publishers.

Contents

About this draft	iii
Exposé XI. Examples and complements	1
1. Projective spaces, unirational varieties	1
2. Abelian varieties	2
3. Projecting cones, Zariski's example	4
4. The exact cohomology sequence	5
5. Special cases of principal bundles	9
6. Application to principal coverings: Kummer and Artin-Schreier theories	10
7. Bibliography	14

EXPOSÉ XI

Examples and complements

1. Projective spaces, unirational varieties

PROPOSITION 1.1. *Let k be an algebraically closed field, $X = \mathbf{P}_k^r$ the projective space of dimension r over k . Then X is simply connected, i.e. $\pi_1(X) = 0$.*

For $r = 0$, this is trivial. If $r = 1$, one must show that if X' is a nonempty connected étale covering of $X = \mathbf{P}_k^1$, then $X' \xrightarrow{\sim} X$. The genus formula gives us here, if g and g' are the genera of X and X' :

$$1 - g' = d(1 - g),$$

where d is the degree of X' over X . Since $g = 0$, we will therefore have $1 - g' = d$, which requires $d = 1$ since $g' \geq 0$, which proves $X' \xrightarrow{\sim} X$. When $r \geq 2$, one proceeds by induction on r , assuming that $\mathbf{P}^{r'}$ is simply connected for $r' < r$. Applying this to a hyperplane of $\mathbf{P}^r$ and using (X 2.10), it indeed follows that $\mathbf{P}^r$ is simply connected. Another proof: we will have $\pi_1(\mathbf{P}^1 \times \cdots \times \mathbf{P}^1) = \pi_1(\mathbf{P}^1) \times \cdots \times \pi_1(\mathbf{P}^1)$ by virtue of (X 1.7), hence $(\mathbf{P}^1)^r$ is simply connected since $\mathbf{P}^1$ is, hence $\mathbf{P}^r$ is simply connected by virtue of the birational invariance of the fundamental group (X 3.4). This proof shows more generally:

COROLLARY 1.2. *Let X be a proper and normal scheme over an algebraically closed field k ; if X is a rational variety, i.e. integral and its function field is a purely transcendental extension of k , then X is simply connected.*

This result applies in particular to Grassmannian varieties and more generally to the varieties G/H , where G is a connected linear group over k and H an algebraic subgroup containing a Borel subgroup of G .

Recall that one calls *unirational variety over k* a proper and integral scheme over k whose function field K is contained in a purely transcendental extension K' of k , finite over K (i.e. having the same transcendence degree over k as K), i.e. if there exists a dominant rational map $f: \mathbf{P}_k^r \rightarrow X$, with $r = \dim X$. If X is normal, one therefore sees by the reflections preceding X 3.4 that for every connected étale covering X' of X , with field L/K , the algebra $L \otimes_K K'$ over K' is unramified over the model $\mathbf{P}^r$, hence completely decomposed by virtue of 1.1, which shows that L is K -isomorphic to a subextension of K'/K . This therefore proves, taking into account V 8.2:

COROLLARY 1.3. *The fundamental group of a normal unirational variety over an algebraically closed field is finite.*

(N.B. One will note that in the definition of “unirational”, one did not need K'/K to be finite).

REMARKS 1.4. Of course, the results of this no. are well known. On the other hand, J-P. Serre has shown [10] that when X is a smooth unirational projective variety over an algebraically closed field of *characteristic zero*, X is simply connected. His proof is transcendental in that it uses the Hodge symmetry theorem.

[Added in 2003 (MR)] Let us restrict ourselves to the case of an algebraically closed field k of characteristic $p \geq 0$. In characteristic $p > 0$, the definition of unirational k -variety given above is that of weakly unirational k -variety, as opposed to that of strongly unirational k -variety, where one assumes moreover that K' is a separable extension of K .

In the case of dimension 2, there exist projective and smooth, weakly unirational surfaces which have a nontrivial fundamental group (finite by 1.3) and which are therefore not rational (T. Shioda, On unirationality of supersingular surfaces, *Math. Ann.* **225** (1977), p. 155–159.). On the other hand, a strongly unirational surface is rational: this follows from Castelnuovo’s rationality criterion, extended to characteristic $p > 0$ by O. Zariski (cf. J.-P. Serre, *Sém. Bourbaki* no. 146, 1957 and *Œuvres—Collected papers*, vol. 1, p. 467).

Over the field $\mathbf{C}$ of complex numbers, one knows examples of smooth projective varieties of dimension ≥ 3 which are unirational and not rational (cf. P. Deligne, *Variétés unirationnelles non rationnelles* [d’après M. Artin et D. Mumford], *Sém. Bourbaki* no. 402, 1971–72, *Lecture Notes* vol. 317). This is the case in particular for a smooth cubic hypersurface in the projective space $\mathbf{P}^4$. Some of these examples extend to characteristic > 0 to give strongly unirational non-rational varieties (cf. J.-P. Murre, Reduction of the proof of the non-rationality of a non singular cubic threefold to a result of Mumford, *Compositio Math.* **27** (1973), p. 63–82).

Let V be a normal, integral projective k -variety. One says that V is rationally connected if there exist an integral k -scheme of finite type T and a k -morphism $F : T \times \mathbf{P}^1 \rightarrow V$ such that the morphism:

$$(1) \quad \begin{aligned} T \times \mathbf{P}^1 \times \mathbf{P}^1 &\longrightarrow V \times V \\ (t, u, u') &\longmapsto (F(t, u), F(t, u')) \end{aligned}$$

is dominant. In particular, through two sufficiently general rational points of V there passes a rational curve. The terminology is justified by the fact that if V is rationally connected, one can join two rational points by a finite chain of rational curves. If k is of characteristic $p > 0$, one strengthens the preceding definition by requiring that the map (1) be generically smooth. One then obtains the notion of separably rationally connected variety. Thus unirational varieties are rationally connected and in characteristic $p > 0$, strongly unirational varieties are separably rationally connected. J. Kollár has shown that separably rationally connected varieties have a trivial algebraic fundamental group (cf. O. Debarre, *Variétés rationnellement connexes* [d’après T. Graber, J. Harris, J. Starr et A.J. de Jong], *Sém. Bourbaki* no. 906, 2001–2002).]

2. Abelian varieties

Let k be an algebraically closed field, A an abelian variety over k , i.e. a group scheme over k , proper over k , smooth over k , and connected, and finally G a commutative group scheme of finite type over k . Let us denote by $\text{Ext}(A, G)$ the group of classes of commutative extensions of A by G , by $H^1(A, G)$ the group of classes of principal bundles on A with group G (compare no. 4 below), and let us consider the canonical homomorphism

$$\text{Ext}(A, G) \longrightarrow H^1(A, G).$$

An argument of Serre [5, Chap. VII, Th. 5] shows that it is an injective homomorphism, whose image is the set of “primitive elements” of $H^1(A, G)$, i.e. of the elements ξ for which one has:

$$\pi^*(\xi) = \text{pr}_1^*(\xi) + \text{pr}_2^*(\xi),$$

where pr_i are the two projections of $A \times A$ onto A , and $\pi : A \times A \rightarrow A$ the composition law of A (N.B. Serre states his theorem only for G linear and connected, and of course smooth over k , but by simplifying the first part of his argument, one sees that these restrictions are unnecessary: it suffices to note that every morphism from A to a group scheme E of finite type over k which takes the unit to the unit is a group homomorphism, and to apply this to the sections over A of an extension E of A by G).

We are going to apply this result to the case where G is a finite group separable over k , i.e. an ordinary finite group, assumed commutative. Using then $\pi_1(A \times A) \xrightarrow{\sim} \pi_1(A) \times \pi_1(A)$ (X 1.7) and interpreting $H^1(X, G)$ as $\text{Hom}(\pi_1(X), G)$ for every algebraic scheme X , in

particular for $X = A$ or $X = A \times A$, one sees that every class of $H^1(A, G)$ is primitive, hence one has an isomorphism

$$\mathrm{Ext}(A, G) \xrightarrow{\sim} H^1(A, G),$$

in other words *every principal covering of A with commutative structure group G , pointed over the origin of A , is endowed in a unique way with a structure of algebraic group admitting the marked point as origin, and such that $A' \rightarrow A$ is a homomorphism of algebraic groups*. In particular, if A' is connected, it is also an abelian variety, isogenous to A .

On the other hand, since the functor $X \mapsto \pi_1(X)$ from pointed algebraic schemes X to groups commutes with products (IX 1.7), it transforms a group in the first category into a group in the category of groups, i.e. into a *commutative* group. Hence *if A is an abelian variety, $\pi_1(A)$ is a commutative group*. Hence, in order to know $\pi_1(A)$, it suffices to know the functor $G \mapsto H^1(A, G) = \mathrm{Hom}(\pi_1(A), G)$ for G varying over the *commutative* finite groups. Finally, let us recall that for every integer $n > 0$, the homomorphism of multiplication by n in A :

$$A \xrightarrow{n} A$$

is surjective, hence with finite kernel, i.e. it is an isogeny, and that it follows that every isogeny $A' \rightarrow A$ is a quotient of an isogeny of the preceding type. From this, and from standard arguments (cf. for example [6]), one derives:

THEOREM 2.1 (Serre-Lang). *Let A be an abelian variety over an algebraically closed field k , and for every integer $n > 0$ let us consider the ordinary finite group K_n underlying the kernel ${}_nA$ of the multiplication by n in A ; finally, let us set, for every prime number ℓ :*

$$T_\ell(A) = \varprojlim_r K_{\ell^r}$$

and

$$T(A) = \prod_\ell T_\ell(A) = \varprojlim_n K_n$$

(where, for m a multiple of n , $m = ns$, one maps K_m into K_n by multiplication by s). Then the group $\pi_1(A)$ is canonically isomorphic to $T(A)$, hence for every prime number ℓ , the ℓ -primary component of $\pi_1(A)$ is canonically isomorphic to $T_\ell(A)$.

One will note that these isomorphisms are functorial for variable A . The module $T_\ell(A)$ is called the ℓ -adic Tate module of the abelian variety A . It is an additive functor in A ; in particular it gives rise to a representation of the ring $\mathrm{Hom}(A, A)$ of endomorphisms of A in $T_\ell(A)$, called the ℓ -adic representation of Weil, and which plays an important role in the theory of abelian varieties (cf. for example [4, Chap VII]). Theorem 2.1 gives an interpretation of it in terms of the natural representation in the ℓ -adic homology group of A , $H_1(A, \mathbf{Z}_\ell) = \pi_1(A)_\ell$, which is evidently more satisfying a priori, notably from the point of view of the Lefschetz formula, [4, Chap V]. Let us recall here Weil's results on the structure of $T_\ell(A)$:

- a) If n is prime to $\mathrm{char}(k)$, then K_n is a free module of rank equal to $2 \dim A$ over $\mathbf{Z}/n\mathbf{Z}$, hence if ℓ is a prime number $\neq \mathrm{char}(k)$, $T_\ell(A)$ is a free module of rank equal to $2 \dim A$ over the ring $\mathbf{Z}_\ell$ of ℓ -adic integers;
- b) If n is a power of $\mathrm{char}(k) = p$, then K_n is a free module of rank $\nu \leq \dim A$ over $\mathbf{Z}/n\mathbf{Z}$, ν independent of n , hence $T_p(A)$ is a free module of rank $\nu \leq \dim A$ over the ring $\mathbf{Z}_p$ of p -adic integers.

This shows that in the theory of the fundamental group developed here, the fundamental group of a variable abelian variety does not vary in a regular way with the parameter, its p -primary component being liable to decrease abruptly for values of the parameter t corresponding to a residue characteristic p ; the best-known case of this phenomenon is that of elliptic curves. One will note, however, that whatever n may be (prime or not to the characteristic), the true kernel ${}_nA$ in A for the multiplication by n is a finite group scheme over k of degree n^{2g} , where $g = \dim A$, which will be non-separable over k if n is a multiple of $p = \mathrm{char}(k)$. Moreover, when A varies in a family of abelian varieties, i.e. if one has

an abelian scheme A over a base scheme S , one shows more generally that ${}_n A$ is a finite and flat group scheme over S , of degree n^{2g} over S ; that is to say, provided one takes into account the infinitesimal parts of the kernels ${}_n A$, they behave in a regular way whatever n may be. This suggests that the “true” fundamental group of an abelian variety A is the algebraic pro-group (formal projective limit of finite groups over k) $\varprojlim_n {}_n A$, where by “true fundamental group” of an algebraic scheme X , one must understand: the pro-group which classifies the principal coverings of X with structure group an arbitrary finite group G over k (not necessarily separable over k). In this way, for example, one recovers, through the representations of $\mathrm{Hom}(A, A)$ in the p -primary component of the true fundamental group of A , the characteristic polynomial of Weil defined by the latter with the help of the $\ell \neq p$, in a more natural way than Serre’s construction [8].

3. Projecting cones, Zariski’s example

As before, let k be an algebraically closed field for simplicity, and let V be a connected projective k -scheme, a closed subscheme of $\mathbf{P}_k^r$, which one may, if one wishes, assume nonsingular. Let $Y = \widehat{C}$ be the projective projecting cone of V , y_0 its vertex, $X = \widehat{C}_V$ the usual projective closure of the vector bundle $C_V = \mathbf{V}(\mathcal{O}_V(1))$ associated with $\mathcal{O}_V(1)$, and finally

$$f: X \longrightarrow Y$$

the canonical morphism, contracting the zero section X_0 of C_V over X to a point (EGA II 8.6.4). Since X is a locally trivial bundle over V , with fibers $\mathbf{P}^1$, hence with simply connected fibers, the morphism $p: X \rightarrow V$ induces, by virtue of (X 1.4), an isomorphism:

$$\pi_1(X) \xrightarrow{\sim} \pi_1(V).$$

Since p induces an isomorphism $X_0 \rightarrow V$, one concludes that *an étale covering of X is completely decomposed if and only if its restriction to X_0 is*. Now for every étale covering Y' of Y , the inverse image $X' = X \times_Y Y'$ is an étale covering of X completely decomposed over the fiber X_0 , hence trivial. Since the homomorphism $\pi_1(X) \rightarrow \pi_1(Y)$ is surjective (IX 3.4), one concludes that

$$\pi_1(Y) = (e)$$

in other words *every projective projecting cone is simply connected*. (N.B. in characteristic 0, the same result will be valid if one takes for Y the affine projecting cone).

Suppose now V regular, i.e. smooth over k ; then X is regular, and for a suitable projective embedding of V , one then finds a *normal* projecting cone Y . If V is not simply connected, hence X not simply connected, let X' be a nontrivial connected étale covering of X . Since the fibers of X at the points $y \in Y$ distinct from y_0 are reduced to a point, one sees that the restriction of X' to its fibers (in particular to the generic fiber) is trivial; however X does not come by inverse image from an étale covering of Y , since Y is simply connected and X' would be completely decomposed. This shows that (X 1.3 and 1.4) become false if one replaces the hypothesis that f is separable by the weaker one that its fibers are separable (or even smooth) algebraic schemes over the $k(s)$. One will note likewise that the fundamental groups of the geometric fibers $\overline{X}_y$ of the $y \neq y_0$ are evidently reduced to (e) since these fibers are reduced to a point, whereas $\pi_1(X_0) \neq e$; hence the semicontinuity theorem (X 2.4) likewise fails for f .

Let us finally indicate the example, pointed out by Zariski, which shows that these same theorems fail when one replaces in them the hypothesis that f is separable by the hypothesis that f is flat. Let $f: X \rightarrow Y$ be a morphism from a nonsingular projective surface to the rational line $Y = \mathbf{P}^1$, such that $K = k(X)$ is a “regular” extension, i.e. primary and separable, of $k(f)$ (i.e. the geometric generic fiber is connected and separable), and such that the divisor $(f) = X_0 - X_\infty$ is an n th multiple of a divisor (where n is an integer prime to the characteristic). It is possible to construct such examples in every characteristic. Let X' be the normalization of X in $K(f^{1/n})$, where $K = k(X)$ is the function field of X . It follows from the hypothesis on (f) that X' is étale over X . Let Y' be the normalization of Y in $k(t)(t^{1/n})$;

it is ramified over Y at the points $t = 0$ and $t = \infty$ exactly, and the restriction $X'|f^{-1}(U)$ is isomorphic to the inverse image of $Y'|U$. In particular, the restriction of X' to the *geometric* generic fiber of X over Y decomposes completely. However, X' is not isomorphic to the inverse image of an étale covering of Y , for one sees at once that the latter would necessarily be Y' , which is absurd since Y' is ramified over Y ¹.

Here is (after Serre) a simple way of realizing the conditions of this example, drawing inspiration from [5, no. 20]: one takes for n a prime number ≥ 5 , distinct from the characteristic, and one lets $G = \mathbf{Z}/n\mathbf{Z}$ act on k^{4*} by multiplying the coordinates by four distinct characters of G (which is possible since $n \geq 5$). Then G acts on the projective space $\mathbf{P}_k^3$, and the only points fixed under G are the four points corresponding to the coordinate axes. The surface X' with equation $x^n + y^n + z^n + t^n = 0$ is smooth over k (Jacobian criterion), and contains none of the fixed points; hence G , being of prime order, acts on X “without fixed points”, i.e. X is a principal covering of $X = X'/G$ with group G . Let $g = x/y$ in $k(X') = K'$; it is a Kummer generator of K' over $K = k(X)$ if the chosen characters were χ^i , $i = 0, 1, 2, 3$, with χ a primitive character; let f be its n th power, which is an element of K . One sees at once that K' is a regular extension of $k(g)$, which follows from the fact that the plane curve with homogeneous equation in U, T, Z : $T^n + Z^n + (1 + g^n)U^n = 0$ is smooth over $k(g)$ (Jacobian criterion), and that one knows that every plane curve is connected. On the other hand one has $k(f) = K \cap k(g)$, since the right-hand side is an extension of $k(f)$ contained in the extension $k(g)$ of prime degree, and distinct from $k(g)$ (since $g \notin K$). This implies that K is a regular extension of $k(f)$. Finally the divisor of f on X is an n th multiple of a divisor, since its inverse image on X' is the divisor of g^n , hence an n th multiple, and one can descend back because X' is étale over X . One would be done if the rational map $f: X \rightarrow \mathbf{P}^1$ were a morphism, that is to say if the divisors of zeros and of poles of f did not meet. In fact, one easily verifies (again by looking on X') that the two divisors in question are the products by n of two curves smooth over k , intersecting transversally at a point a . Replacing now X by the scheme obtained by blowing up a , the preceding conditions ($\text{div}(f)$ divisible by n , and $k(X_1) = k(X)$ regular extension of $k(X)$) remain satisfied, but moreover f is a *morphism* $X_1 \rightarrow \mathbf{P}^1$, hence we are under the desired conditions.

4. The exact cohomology sequence

Let S be a prescheme, so that the category $\mathbf{Sch}/_S$ of preschemes over X is determined, hence also the notion of group in it, which will also be called a *group prescheme over S* , or simply an *S -group*. To simplify the exposition and fix ideas, we shall most often restrict ourselves in what follows to groups that are *affine* and *flat* over S^2 , which will suffice for the applications we have in view. (Of course, one encounters numerous cases where neither the one nor the other hypothesis is satisfied). Let G be such an S -group, and let P be a prescheme over S on which G acts on the right, which implies in particular a morphism

$$\pi : P \times_S G \longrightarrow P$$

in the category $\mathbf{Sch}/_S$, satisfying the well-known axioms. One says that P is *formally principal homogeneous under G* if the morphism

$$P \times_S G \longrightarrow P \times_S P$$

with components pr_1 and π is an isomorphism; it amounts to the same to say that for every object S' of $\mathbf{Sch}/_{S'}$, $P(S') = \text{Hom}_S(S', P)$, considered as a set with groups of operators $G(S') = \text{Hom}_S(S', G)$, is empty or principal homogeneous (i.e. empty or isomorphic to $G(S')$ on which the group $G(S')$ acts by right translations). One says that P is *trivial* if P is

¹One may remark, from the point of view of the “étale topology” (SGA 4 VII), that in this example $R^1(f_{\text{ét}})_*(\mathbf{Z}/n\mathbf{Z})$ is “non-separated” over S .

^{*}[Added in 2003: k^4 denotes the affine space of dimension 4 over k .]

²In fact, for what follows, the quasi-affine hypothesis instead of the affine one would suffice, cf. footnote (5), p. 7.

isomorphic to G , on which G acts by right translations, or what amounts to the same, if each of the sets with operators $P(S')$ under $G(S')$ is trivial. One verifies, for example by the patented procedure of passing to the set-theoretic case, that P is trivial if and only if it is formally principal homogeneous, and admits a section over S (this last fact being stated in categorical terms by saying that P has a section over the final object $e = S$ of $\mathbf{Sch}/_S$, i.e. that there exists a morphism from e to P). To define the notion of principal homogeneous bundle P under G , stronger than that of formally principal homogeneous bundle, one must first specify in $\mathbf{Sch}/_S$ a set of morphisms that will be used for “descent”, and will play the role of “localization morphisms” in order to “trivialize” bundles. The most suitable choice varies with the context, none containing all the others³. Here it will be convenient to adopt the following definition:

DEFINITION 4.1. Let G be an S -group. One calls a *principal homogeneous bundle* (on the right) under G an S -prescheme P with S -group G acting on the right, such that there exist a covering of S by open sets U_i , and for every i a faithfully flat and quasi-compact change of base morphism $S'_i \rightarrow U_i$, such that $P' = P \times_S S'$ is a trivial prescheme with operators under $G' = G \times_S S'$, where S' is the S -prescheme disjoint sum of the S'_i .

(Note that the change of base functor $X \mapsto X' = X \times_S S'$, being left exact, transforms groups into groups, objects with group of operators into objects with groups of operators). Let us note that 4.1 is *stable under change of base*. Let us also note:

PROPOSITION 4.2. Let G be an S -group, flat and quasi-compact over S , and P an S -prescheme on which G acts on the right. The following conditions are equivalent:

- (i) P is a principal homogeneous bundle under G ;
- (ii) P is formally principal homogeneous under G , and the structure morphism $P \rightarrow S$ is faithfully flat and quasi-compact.

If P is principal homogeneous under G , then with the notation of 4.1, P' is faithfully flat and quasi-compact over S' (since G' is, and P' is S' -isomorphic to it), hence P has the same properties over S (for “surjective” and “quasi-compact”, cf. VIII 3.1; for “flat” it is an omission in the sorites of Exposé VIII). Conversely, if (ii) is satisfied, let us take the change of base $S' = P$, which is indeed faithfully flat and quasi-compact over S ; then P' will be formally principal homogeneous over S' since P is so over S and the change of base functor is left exact; on the other hand P' has a section over S' , namely the diagonal section, hence it is a trivial principal bundle, which completes the proof.

COROLLARY 4.3. If G is affine and flat over S , every principal homogeneous bundle P under G is affine and flat over S .

Indeed, it becomes so after a faithfully flat and quasi-compact extension of the base, and one applies (VIII 5.6).

The usefulness of Definition 4.1 for S -groups that are *flat* and *affine* over S is due to (VIII 2.1), i.e. to the fact that the morphisms $S' \rightarrow S$ considered in 4.1 are effective descent morphisms for the category of preschemes affine over others. Thanks to this fact, the verification of the facts sketched below is done in an essentially “categorical” way⁴. Let E be an S -prescheme on which the S -group G acts on the left, and let P be a principal homogeneous bundle (on the right) under G ; we want to define an associated bundle $E^{(P)}$, “locally” isomorphic to E . For this, let us make G act on the right on $P \times_S E$ according to the law $(x, y) \mapsto (xg, g^{-1}y)$, which describes such operations in the set-theoretic context, and extends to categories by the patented procedure. We shall set, subject to existence:

$$E^{(P)} = (P \times_S E)/G$$

³See on this subject SGA 3 IV, in particular §4.

⁴Cf. loc. cit. in footnote (3), p. 6.

whereupon one finds that $P \times_S E$ will be a prescheme over $T = E^{(P)}$, with right groups of operators $G_T = G \times_S T$; to be at ease, one would like moreover $P \times_S E$ to be a principal homogeneous bundle over T with group G_T . To verify the existence of $E^{(P)}$ and the preceding property, let us take again the S' of Definition 4.1 and look at the situation over S' that is the inverse image of the initial situation. From the fact that P' is trivial, i.e. isomorphic to G'_d , one sees at once that $E'^{(P')}$ exists, and has the desired exactness property. In fact, $E' \times_{S'} P'$ is G' -isomorphic to the product $E' \times_{S'} G'$, hence $E'^{(P')}$ is isomorphic to E' . Moreover, the formation of the “associated bundle” in the case of a trivial space with operators commutes with every extension of the base, and taking in this instance the various extensions of the base $S'' \rightrightarrows S'$ and $S''' \rightrightarrows S'$, where S'' and S''' are the double and triple fiber products of S' over S , one finds that $E'^{(P')}$ is endowed with a descent datum relative to the morphism $S' \rightarrow S$, and that $E^{(P)}$ exists with the required properties if and only if this descent datum is effective; of course $E^{(P)}$ is then nothing but the descended object. (Use the fact that $S' \rightarrow S$ is a descent morphism in the category of S -preschemes, cf. VIII 5.2). It follows that *the associated bundle exists if E is affine over S* . We shall apply this construction to the case where one has a homomorphism of S -groups $G \rightarrow H$, and where one takes for E the S -prescheme H endowed with the left operations of G on H resulting from the given morphism; since H acts on the right on itself in such a way as to commute with the operations of G on H , and since (subject to existence over S) the formation of the associated bundle commutes with extension of the base, one easily finds that H will act on the right on $P^{(H)}$, which is therefore a principal homogeneous bundle under H in the sense of 4.1, and more precisely is trivialized by the same morphism $S' \rightarrow S$ as P . In particular, *to every principal homogeneous bundle P under G and every homomorphism of S -groups $G \rightarrow H$, with H affine over S , is associated a principal homogeneous bundle with group H , in a way functorial in $(G \rightarrow H)$, and compatible with arbitrary changes of base $T \rightarrow S$.*

DEFINITION 4.4. Let G be an S -prescheme. One denotes by $H^0(S, G)$ the set of sections of G over S , which one will regard as a group when G is an S -group. In this case, one denotes by $H^1(S, G)$ the set of classes, up to isomorphism, of principal homogeneous bundles over S with group S , regarding $H^1(S, G)$ as endowed with the “marked point” that corresponds to the trivial bundles⁵.

Thus, $H^0(S, G)$ is a functor in the S -prescheme G , with values in the category of sets. This functor is left exact, a fortiori commutes with finite products, which indeed implies that it transforms groups into groups, commutative groups into commutative groups. Similarly, $H^1(S, G)$ is a functor in the *affine* S -group G , with values in the category of sets, thanks to the formation of associated bundles; one easily finds that this functor commutes with finite products. In particular it transforms the groups in the category of affine S -groups, i.e. the *commutative affine* S -groups, into groups of the second, and even into commutative groups (since the groups of the first category are commutative). Thus, *if G is a commutative affine S -group, $H^1(S, G)$ is a commutative group*, and a homomorphism $G \rightarrow H$ of commutative affine S -groups gives rise to a homomorphism of groups $H^1(S, G) \rightarrow H^1(S, H)$.

For simplicity, we restrict ourselves in what follows to the consideration of *affine and commutative* S -groups. Let

$$0 \longrightarrow G' \xrightarrow{u} G \xrightarrow{v} G'' \longrightarrow 0$$

be a sequence of morphisms of such groups; *we shall say that this sequence is exact if $vu = 0$ (which allows one to regard G as a prescheme over G'' , with right groups of operators $G'_{G''}$) and if G is a principal homogeneous bundle over G'' with group $G'_{G''} = G' \times_S G''$* . This implies in particular that $u: G' \rightarrow G$ is a kernel of v , and a fortiori it implies the exactness of the

⁵This notation is consistent with the general cohomological notation (SGA 4 V) only when one has criteria for effectiveness of descent, which are hardly ensured except if G is affine (or only quasi-affine, cf. (VIII 7.9)).

sequence

$$0 \longrightarrow H^0(X, G') \longrightarrow H^0(X, G) \longrightarrow H^0(X, G'').$$

It implies moreover the possibility of defining a map

$$\partial: H^0(X, G'') \longrightarrow H^1(X, G'),$$

by associating to every section of G'' over S , i.e. to every S -morphism $f: S \rightarrow G''$, the principal homogeneous bundle P_f with group $G' \simeq f^*(G'_{G''})$ over S , inverse image of the principal homogeneous bundle G' over G'' . From the point of view of S -preschemes, it is therefore nothing but the inverse image under $v: G \rightarrow G''$ of the subprescheme image of S under the immersion f , and the operations of G' on P_f are induced by the right operations of G' on G . We likewise leave to the reader the verification of the following proposition, which presents no difficulties other than those of writing it up:

PROPOSITION 4.5. *The map $\partial: H^0(X, G'') \rightarrow H^1(X, G')$ is a homomorphism of groups. The following sequence of homomorphisms is exact:*

$$\begin{aligned} 0 \longrightarrow H^0(X, G') \xrightarrow{u^0} H^0(X, G) \xrightarrow{v^0} H^0(X, G'') \xrightarrow{\partial} H^1(X, G') \\ \xrightarrow{u^1} H^1(X, G) \xrightarrow{v^1} H^1(X, G'') \end{aligned}$$

(where the homomorphisms other than ∂ come from the functoriality of H^0 resp. H^1).

REMARKS 4.6. The point of view presented here for the study of principal homogeneous bundles is visibly inspired by Serre [7], which the reader will have every interest in consulting. When one wants a formalism that applies also to structural S -groups that are quasi-projective over S (so as to include projective abelian schemes in particular), it is advantageous to modify 4.1 by requiring there that S' be a sum of preschemes S'_i that are finite and locally free over open sets S_i of S covering S . The preceding developments are then valid, including in particular 4.5, on replacing everywhere the affine hypothesis by the quasi-projective hypothesis, and on interpreting correspondingly the definition given above of an exact sequence of S -groups. It suffices indeed to replace the reference to (VIII 2.1) by (VIII 7.7): the morphisms $S' \rightarrow S$ used are effective descent morphisms for the fibered category of preschemes quasi-projective over others. One should take care, however, that this second notion of principal homogeneous bundle is more restrictive than the first 4.1.

4.7. One obtains an even more restrictive notion of principal homogeneous bundle by requiring that S be covered by open sets S_i such that for every i , $P|_{S_i}$ is a trivial bundle with operators under $G|_{S_i}$: one will then say that P is a *locally trivial* principal homogeneous bundle. The classes of these bundles, for given G , form a subset of $H^1(X, G)$, which is in one-to-one correspondence with $H^1(X, \mathcal{O}_X(G))$, where $\mathcal{O}_X(G)$ is the sheaf (in the ordinary sense) of sections of G over S , cf. [2]. For these H^1 to still give rise to an exact cohomology sequence 4.5, one must obviously assume that the sequence $0 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 0$ is exact in the reasonable sense for this new context, i.e. that G is a locally trivial bundle over G'' , with group $G'_{G''}$; this also means that $u: G' \rightarrow G$ is a kernel of $v: G \rightarrow G''$, and that G locally admits a section over G'' .

4.8. It is obviously very desirable to continue the exact sequence 4.5 by introducing the higher cohomology groups $H^i(X, G)$. This is possible by placing oneself in the framework of “Weil cohomology”: one considers the category $\mathcal{B}$ of quasi-compact preschemes over S , endowed with the set $\mathcal{M}$ of faithfully flat and quasi-compact morphisms, which will be called localizing morphisms. An abelian “Weil sheaf” on S (or better, on $(\mathcal{B}, \mathcal{M})$) is then a contravariant functor $\mathcal{F}$ from $\mathcal{B}$ to the category of abelian groups, transforming sums into products, and a sequence $T'' = T' \times_T T' \xrightarrow{\text{pr}_1, \text{pr}_2} T' \xrightarrow{f} T$, with $f \in \mathcal{M}$, into an *exact* diagram of sets $\mathcal{F}(T) \longrightarrow \mathcal{F}(T') \rightrightarrows \mathcal{F}(T'')$. The Weil sheaves form an abelian category

with exact inductive limits admitting a generator, hence admitting enough injective objects [1]. The right derived functors of the functor $\Gamma(\mathcal{F}) = \mathcal{F}(S)$ are then denoted $H^i(S, \mathcal{F})$. On the other hand, every commutative S -group obviously defines a Weil sheaf (VIII 5.2), whose H^0 and H^1 are none other than $H^0(S, G)$ and $H^1(S, G)$, which allows one to define the other $H^i(S, G)$ in a reasonable way. One shows moreover that an exact sequence of S -groups defines an exact sequence of Weil sheaves, which allows one to recover and to extend the exact sequence 4.5⁶.

4.9. It would be advisable to develop the noncommutative variants of 4.5 as in [2]. For a systematic development, in the appropriate framework, of the various cohomological notions sketched in the present no., we refer to a work in preparation by J. GIRAUD⁷.

5. Special cases of principal bundles

Suppose now that S is connected and endowed with a geometric point a , whence a fundamental group $\pi_1(S, a)$ making it possible to classify the étale coverings of S : the category of étale coverings of S is equivalent to the category of finite sets on which π_1 acts continuously. It follows that a finite and étale group scheme G over S is essentially determined by an ordinary finite group $\mathcal{G}$, on which π_1 acts continuously by group automorphisms. An étale covering P of S on which G acts on the right is essentially determined by a finite set $\mathcal{P}$ on which π_1 acts continuously (on the left), and on which $\mathcal{G}$ acts on the right in a way compatible with the operations of π_1 :

$$s(p \cdot g) = (sp) \cdot (sg) \quad \text{for } s \in \pi_1, p \in \mathcal{P}, g \in \mathcal{G}.$$

One verifies that P is a principal homogeneous bundle in the sense of 4.1 if and only if $\mathcal{P}$ is a principal homogeneous set under $\mathcal{G}$ (use for example criterion 4.2). In other words, *the category of principal homogeneous bundles over S with group G is equivalent to the category of principal homogeneous bundles with group G in the category of finite sets on which π_1 acts continuously*. One deduces from this in particular a canonical bijection, functorial in G :

$$(*) \quad H^1(S, G) \xrightarrow{\sim} H^1(\pi_1, \mathcal{G}),$$

where the right-hand side denotes the set of classes, up to isomorphism, of principal homogeneous bundles under $\mathcal{G}$ in the category of finite sets on which π_1 acts (there is moreover no need to specify: continuously), a set which is made explicit in a well-known way as the quotient set of the set $Z^1(\pi_1, \mathcal{G})$ of 1-cocycles $\varphi : \pi_1 \rightarrow \mathcal{G}$ (satisfying $\varphi(1) = 1$, $\varphi(st) = \varphi(s)(s.\varphi(t))$) by the group $\mathcal{G}$, which acts on it in a natural way).

An important case is that where π_1 acts trivially on $\mathcal{G}$, i.e. when G is a completely decomposed covering of S , isomorphic to the sum of $\mathcal{G}$ copies of S ; one then also writes $H^1(S, \mathcal{G})$ instead of $H^1(S, G)$, and this set is in one-to-one correspondence $(*)$ with $H^1(\pi_1, \mathcal{G}) = \text{Hom}(\pi_1, \mathcal{G})/\text{inner automorphisms of } \mathcal{G}$. Note moreover that in this case, a principal homogeneous bundle over S with group G is nothing but a *principal covering* of S with group $\mathcal{G}$ (V 2.7), and the preceding one-to-one correspondence is the one deduced from the correspondence between principal coverings of S with group $\mathcal{G}$, *pointed* over a , and the continuous homomorphisms from $\pi_1(S, a)$ to $\mathcal{G}$ (V, end of no. 5).

The interest of relating the theory of étale coverings with that of principal bundles (already implicit in A. WEIL, Généralisation des Fonctions Abéliennes, and made explicit by S. LANG in his geometric class field theory, cf. Serre [5]) comes from the fact that every S -group that is finite and étale over S can be embedded in an S -group H , affine and smooth over S , with connected fibers, commutative when G is, so that by the exact sequence 4.5 (and possibly its noncommutative variants), the “discrete” classification of principal coverings with group G can be studied with the help of the “continuous” classification of principal bundles with group H , and conversely, for that matter. For the idea of the general construction of the

⁶For a systematic study of this point of view, cf. SGA 4 I to IX.

⁷Cf. J. GIRAUD, *Cohomologie non abélienne*, Springer-Verlag 1971.

immersion of G into H (rather little used in practice, it seems), refer to [5, VI 2.8]. We content ourselves with developing in the next no. two important special cases, which are moreover classical. We shall need there an auxiliary result:

PROPOSITION 5.1. *Let S be a prescheme, and G an S -group isomorphic to $\mathbf{GL}(n)_S$ (for example $\mathbf{G}_{mS}$) or to $\mathbf{G}_{aS}$; then every principal homogeneous bundle under G is locally trivial.*

Let us make precise that $\mathbf{GL}(n)_S$ (n an integer ≥ 0) denotes the S -group that represents the contravariant functor $T \mapsto \mathbf{GL}(n, \Gamma(T, \mathcal{O}_T))$ in the S -prescheme T ; in particular $\mathbf{G}_{mS}$ (“multiplicative group over S ”) represents the contravariant functor $T \mapsto \Gamma(T, \mathcal{O}_T^*)$, hence as a prescheme over S is isomorphic to $\mathrm{Spec} \mathcal{O}_S[t, t^{-1}]$, where t is an indeterminate. Similarly $\mathbf{G}_{aS}$ represents the contravariant functor $T \mapsto \Gamma(T, \mathcal{O}_T)$; it is therefore isomorphic as an S -prescheme to $\mathrm{Spec}(\mathcal{O}_S[t])$, where t is an indeterminate. Note that by dévissage, 5.1 gives back the local triviality result of Rosenlicht, relating to the case where G admits a “composition series” whose consecutive factors are groups of the type considered here. (For a finer study of the questions of local triviality of principal homogeneous bundles, cf. [7] and [3]).

The first assertion is proved by remarking that $G(T) = \mathrm{Aut}(\mathcal{O}_T^n)$, and that the morphisms $S' \rightarrow S$ occurring in 4.1 (i.e. which are faithfully flat and quasi-compact) are effective descent morphisms for the fibered category of Modules locally isomorphic to $\mathcal{O}_T^n$, i.e. locally free of rank n (VIII 1.12). The second is proved in an analogous way, by noting that in this case one has $G(T) = \mathrm{Aut}(\mathcal{E}_T)$, where $\mathcal{E}_T$ is the trivial *extension* of $\mathcal{O}_T$ by $\mathcal{O}_T$ (and where the automorphisms must of course respect the extension structure), that the morphisms $S' \rightarrow S$ occurring in 4.1 are effective descent morphisms for the fibered category of extensions of $\mathcal{O}_T$ by $\mathcal{O}_T$ (as follows easily from VIII 1.1), and that such extensions are automatically locally trivial.

REMARK 5.2. Note that the same type of proof applies to the symplectic group $\mathbf{Sp}(2n)_S$, taking into account that an alternating form on a module locally isomorphic to $\mathcal{O}_S^{2n}$ which is “nondegenerate”, i.e. defines an isomorphism of this Module onto its dual, is locally isomorphic to the standard form. The result for the orthogonal group is on the other hand false, already if S is the spectrum of a field, because there can be quadratic forms over a field that are not isomorphic to the standard form. Moreover, one essentially shows in [3] that the groups $\mathbf{GL}$, $\mathbf{Sp}$, $\mathbf{G}_a$ and those that decompose by dévissage into such groups are, more or less, the only ones for which one has a local triviality result of the type considered here.

COROLLARY 5.3. *One has canonical bijections*

$$H^1(S, \mathbf{GL}(n)_S) \xleftarrow{\sim} H^1(S, \mathbf{GL}(n, \mathcal{O}_S)),$$

in particular

$$H^1(S, \mathbf{G}_{mS}) \xleftarrow{\sim} H^1(S, \mathcal{O}_S^*),$$

and

$$H^1(S, \mathbf{G}_{aS}) \xleftarrow{\sim} H^1(S, \mathcal{O}_S),$$

where the right-hand sides denote cohomologies of the topological space S with coefficients in ordinary sheaves.

In particular, $H^1(S, \mathbf{GL}(n)_S)$ is identified with the set of classes, up to isomorphism, of locally free Modules of rank n over S , and $H^1(S, \mathbf{G}_{aS})$ is identified with the set of classes of extensions of the Module $\mathcal{O}_S$ by itself.

6. Application to principal coverings: Kummer and Artin-Schreier theories

PROPOSITION 6.1. *Let S be a prescheme, n an integer > 0 , let $u_n: \mathbf{G}_{mS} \rightarrow \mathbf{G}_{mS}$ be the n th power homomorphism, and μ_n its kernel. Then μ_n is finite and locally free of rank n over S , and it is étale over S if and only if for every $s \in S$, the characteristic of s is prime to n . The sequence of homomorphisms*

$$0 \longrightarrow \mu_n \longrightarrow \mathbf{G}_{mS} \xrightarrow{u_n} \mathbf{G}_{mS} \longrightarrow 0$$

is exact in the sense of no. 4. (It will be called the Kummer exact sequence over S , relative to the integer n).

We have

$$\mathbf{G}_m = \operatorname{Spec} \mathcal{O}_S[t, t^{-1}],$$

and u_n corresponds to the homomorphism u_n on the affine $\mathcal{O}_S$ -algebras, given by

$$u_n(t) = t^n,$$

on the other hand the unit section of $\mathbf{G}_{mS}$ corresponds to the augmentation homomorphism of $\mathcal{O}_S$ -algebras, given by

$$\varepsilon(t) = 1,$$

whose kernel is therefore the principal Ideal $(t - 1)$. The image of the latter under u_n is therefore the principal Ideal $(1 - t^n)$, and one finds:

$$\mu_{nS} = \operatorname{Spec} \mathcal{O}_S[t]/(1 - t^n),$$

which shows in particular that μ_{nS} is finite over S , and defined by an Algebra over S which is free of rank n , having the basis formed by the t^i ($0 \leq i \leq n - 1$). For it to be étale at $s \in S$, it is necessary and sufficient that the reduced Algebra $k[t]/(1 - t^n)$, where $k = k(s)$, obtained by formal adjunction of the n th roots of unity to k , be separable over k , i.e. the roots of $1 - t^n$ in an algebraic closure of k are all distinct, which is equivalent to the fact that n is prime to the characteristic. Finally, to show that the sequence of homomorphisms in 6.1 is exact, one is reduced, by virtue of criterion 4.2, to proving that u_n is faithfully flat. One can obviously assume S affine with ring A , hence $\mathbf{G}_{mS}$ affine with ring $B = A[t, t^{-1}]$, and it suffices to verify that u_n makes B a free module of rank n over A , or, what amounts to the same, that u_n is injective, and that $A[t, t^{-1}]$ is a free module of rank n over $A[t^n, t^{-n}]$. Indeed, one easily verifies that the t^i ($0 \leq i \leq n - 1$) form a basis of the one over the other, which completes the proof.

DEFINITION 6.2. We call μ_{nS} the *Kummer group of rank n over S* , and we call *Kummer principal covering of rank n over S* every principal homogeneous bundle over S with group the Kummer group of rank n .

The set of these coverings is a group, denoted $H^1(S, \mu_{nS})$ or simply $H^1(S, \mu_n)$. Note that the formation of the Kummer group of rank n over S is compatible with base extension, hence that μ_{nS} comes by base extension from the *absolute Kummer group* μ_n over $\operatorname{Spec}(\mathbf{Z})$.

Let us denote by $(\mathbf{Z}/n\mathbf{Z})_S$ the S -group defined by the ordinary finite group $\mathbf{Z}/n\mathbf{Z}$. If G is an arbitrary S -group, the homomorphisms of S -groups u from $(\mathbf{Z}/n\mathbf{Z})_S$ to G correspond one-to-one, and compatibly with change of base, to the sections of G over S whose n th power is the unit section, by associating with u the image under u of the section of $(\mathbf{Z}/n\mathbf{Z})_S$ over S defined by the generator 1 mod $n\mathbf{Z}$ of $\mathbf{Z}/n\mathbf{Z}$. This being said:

COROLLARY 6.3. *If μ_{nS} is étale over S , one thus obtains a one-to-one correspondence between the isomorphisms of S -groups $(\mathbf{Z}/n\mathbf{Z})_S \xrightarrow{\sim} \mu_{nS}$, and the sections of $\mathcal{O}_S$ which are of order exactly n on each connected component of S (such a section will be called a “primitive n th root of unity over S ”). Hence for μ_{nS} to be isomorphic as an S -group to $(\mathbf{Z}/n\mathbf{Z})_S$, it is necessary and sufficient that it be étale over S , i.e. that the residue characteristics of S be prime to n , and that there exist a primitive n th root of unity over S .*

This explains the role played in classical Kummer theory by the hypothesis that the base field (playing the role of S) be of characteristic prime to n and contain the n th roots of unity, and by the choice of a primitive n th root of unity. Once one has the language of schemes at one's disposal, there is no longer any reason to encumber oneself with these hypotheses, and one should reason directly on μ_n instead of $\mathbf{Z}/n\mathbf{Z}$. Thus, the conjunction of 6.1, 4.5 and 5.3 gives us the following general relation between the theory of Kummer principal coverings and that of Picard groups:

PROPOSITION 6.4. *Let S be a prescheme; one has a canonical exact sequence*

$$0 \rightarrow H^0(S, \mu_n) \rightarrow H^0(S, \mathcal{O}_S^*) \rightarrow H^0(S, \mathcal{O}_S^*) \rightarrow H^1(S, \mu_n) \rightarrow H^1(S, \mathcal{O}_S^*) \rightarrow H^1(S, \mathcal{O}_S^*),$$

hence, setting $H^1(S, \mathcal{O}_S^) = \text{Pic}(S)$, and denoting, for every abelian group A , by ${}_nA$ and A_n the kernel and cokernel of multiplication by n in A , the exact sequence:*

$$0 \rightarrow H^0(S, \mathcal{O}_S^*)_n \rightarrow H^1(S, \mu_n) \rightarrow {}_n\text{Pic}(S) \rightarrow 0.$$

We are going to make explicit two important cases, where one or the other extreme term of this exact sequence is zero:

COROLLARY 6.5. *Suppose ${}_n\text{Pic}(S) = 0$, (for example that S is the spectrum of a local ring, or of a factorial ring), and let A be the ring $H^0(S, \mathcal{O}_S)$. Then one has a canonical isomorphism*

$$H^1(S, \mu_n) \xrightarrow{\sim} A^*/A^{*n}.$$

This is essentially the classical statement of Kummer theory, when S is the spectrum of a field.

COROLLARY 6.6. *Suppose that every element of $H^0(S, \mathcal{O}_S)$ is an n th power, for example that $H^0(S, \mathcal{O}_S)$ is a composite of algebraically closed fields or that S is reduced and proper over an algebraically closed field k . Then one has a canonical isomorphism*

$$H^1(S, \mu_n) \xrightarrow{\sim} {}_n\text{Pic}(S).$$

In particular, when S is proper and connected over an algebraically closed field k , this relates the fundamental group of S to the points of finite order of the Picard scheme P of S over k ; thus one will have an isomorphism

$$\text{Hom}(\pi_1(S), \mathbf{Z}/n\mathbf{Z}) \simeq {}_nP(k)$$

for n prime to the characteristic, which is often used in algebraic geometry. As an application, when the connected component P^0 of P is a complete group scheme, of dimension g , one sees, using the results recalled in no. 2 and the finiteness of the torsion group of Néron-Severi, that for every prime number ℓ prime to the characteristic, the ℓ -primary component of the fundamental group $\pi_1(S)$ made abelian is a module of finite type and of rank $2g$ over the ring $\mathbf{Z}_\ell$ of ℓ -adic integers (and moreover free except for at most a finite number of values of ℓ). As Serre has remarked, this makes it possible to prove, under certain conditions, that when X is a flat and projective scheme over a connected S , then the Picard schemes of the fibers of X all have the same dimension, by applying the semicontinuity theorem (X 2.3); Serre's argument applies as soon as the Picard scheme of X over S exists and the connected Picards of the fibers of X over S are proper group schemes, for example when the geometric fibers of X over S are normal (X still being flat and projective over S), in particular if X is smooth and projective over S .

Now let p be a prime number, and suppose that S is a prescheme of characteristic p , i.e. such that $p \cdot \mathcal{O}_S = 0$. Then the p th power homomorphism in $\mathcal{O}_S$ is additive, and the corresponding morphism, obtained by replacing S by a variable T over S :

$$F: \mathbf{G}_{aS} \rightarrow \mathbf{G}_{aS}$$

is therefore a homomorphism of S -groups, which is called the *Frobenius homomorphism* (N.B. Such a morphism is defined for every S -prescheme G which comes by base extension from a prescheme G_0 over the prime field $\mathbf{Z}/p\mathbf{Z}$, and this morphism is a group homomorphism if G_0 is a group prescheme). We shall set:

$$\varphi = \text{id} - F: \mathbf{G}_{aS} \rightarrow \mathbf{G}_{aS}.$$

Consider on the other hand the S -group $(\mathbf{Z}/p\mathbf{Z})_S$ defined by the ordinary finite group $\mathbf{Z}/p\mathbf{Z}$; we have said that for every S -group G , the homomorphisms of S -groups from $(\mathbf{Z}/p\mathbf{Z})_S$ to G correspond one-to-one to the sections of G over S whose p th power is the unit section. When $G = \mathbf{G}_{aS}$, they therefore correspond to arbitrary sections of G over S . Taking in particular

the section of $\mathbf{G}_{aS}$ over S corresponding to the unit section of the sheaf of rings $\mathcal{O}_S$, one finds a homomorphism of S -groups

$$i: (\mathbf{Z}/p\mathbf{Z})_S \longrightarrow \mathbf{G}_{aS}$$

PROPOSITION 6.7. *The sequence of homomorphisms of S -groups*

$$0 \longrightarrow (\mathbf{Z}/p\mathbf{Z})_S \longrightarrow \mathbf{G}_{aS} \longrightarrow \mathbf{G}_{aS} \longrightarrow 0$$

is exact (in the sense of no. 4). (It is called the Artin-Schreier exact sequence over S).

It suffices to prove it over the prime field $k = \mathbf{Z}/p\mathbf{Z}$. It suffices to note that the homomorphism $\wp^*: k[t] \rightarrow k[t]$ defined by $\wp^*(t) = t - t^p$ makes $k[t]$ a free module of rank p over $k[t]$, more precisely that $k[t]$ is a free module over $k[s]$, where $s = t - t^p$, having the basis formed by the t^i ($0 \leq i \leq p-1$).

One concludes from this, using 4.5 and 5.3:

PROPOSITION 6.8. *One has a canonical exact sequence:*

$$0 \longrightarrow H^0(S, \mathbf{Z}/p\mathbf{Z}) \longrightarrow H^0(S, \mathcal{O}_S) \longrightarrow H^0(S, \mathcal{O}_S) \longrightarrow H^1(S, \mathbf{Z}/p\mathbf{Z}) \\ \longrightarrow H^1(S, \mathcal{O}_S) \longrightarrow H^1(S, \mathcal{O}_S),$$

hence an exact sequence:

$$0 \longrightarrow H^0(S, \mathcal{O}_S) / \wp H^0(S, \mathcal{O}_S) \longrightarrow H^1(S, \mathbf{Z}/p\mathbf{Z}) \longrightarrow H^1(S, \mathcal{O}_S)^F \longrightarrow 0,$$

(where the exponent F in the last term means the subgroup of invariants under the endomorphism F , equal to the kernel of $\wp = \text{id} - F$).

Let us make explicit two more extreme cases:

COROLLARY 6.9. *Suppose that $H^1(S, \mathcal{O}_S)^F = 0$, for example that S is an affine scheme. Then, setting $A = H^0(S, \mathcal{O}_S)$, one has a canonical isomorphism*

$$H^1(S, \mathbf{Z}/p\mathbf{Z}) \xrightarrow{\sim} A / \wp A.$$

This is *Artin-Schreier theory* in the classical form, at least when A is the spectrum of a field.

COROLLARY 6.10. *Suppose that $\wp H^0(S, \mathcal{O}_S) = H^0(S, \mathcal{O}_S)$, for example that $H^0(S, \mathcal{O}_S)$ is a composite of algebraically closed fields, or that S is proper over an algebraically closed field. Then one has a canonical isomorphism:*

$$H^1(S, \mathbf{Z}/p\mathbf{Z}) \xrightarrow{\sim} H^1(S, \mathcal{O}_S)^F$$

REMARKS 6.11. The last statement is due to J-P. Serre [9]. It is also possible to develop an analogous theory for the structure group $\mathbf{Z}/p^n\mathbf{Z}$ for arbitrary n , using instead of $\mathbf{G}_a$ the Witt group scheme $\mathbf{W}_n$, cf. loc. cit. Note that in characteristic $p > 0$, Kummer theory no longer gives any information on principal coverings of order p , since μ_p is then an “infinitesimal” group i.e. radicial over the base, hence without direct relation to $\mathbf{Z}/p\mathbf{Z}$; so at first sight, the theory of these coverings is no longer amenable (when S is a proper scheme over an algebraically closed field, to fix ideas) to the theory of the Picard scheme as in 6.6. Nevertheless, if one recalls that the Zariski tangent space at the origin in $\mathbf{Pic}_{S/K}$ ⁸ is identified with $H^1(S, \mathcal{O}_S)$, one observes that *knowledge of the group scheme ${}_p\mathbf{Pic}_{S/k}$, kernel of multiplication by p in $\mathbf{Pic}_{S/k}$, implies that of $H^1(S, \mathbf{Z}/p\mathbf{Z})$ as well as that of $H^1(S, \mu_p)$; note that it also implies that of $H^1(S, \alpha_p)$, where α_p denotes the infinitesimal group scheme over the prime field, kernel of $F: \mathbf{G}_a \rightarrow \mathbf{G}_a$ (which can also be described as the spectrum of the restricted enveloping algebra of the trivial p -Lie algebra of dimension 1): indeed the exact sequence 4.5 gives here:*

$$H^1(S, \alpha_p) \simeq \text{Ker}(F: H^1(S, \mathcal{O}_S) \longrightarrow H^1(S, \mathcal{O}_S)),$$

⁸For the definition of $\mathbf{Pic}_{S/K}$, cf. A. Grothendieck, Sém. Bourbaki no. 232 (February 1962).

and more generally, denoting by α_{p^n} the kernel in $\mathbf{G}_a$ of the n th iterate of F , one will have

$$H^1(S, \alpha_{p^n}) \simeq \text{Ker}(F^n: H^1(S, \mathcal{O}_S) \longrightarrow H^1(S, \mathcal{O}_S)).$$

In fact, knowledge of ${}_p\mathbf{Pic}_{S/k}$ is equivalent to that of $H^1(S, G)$ for every finite commutative algebraic group annihilated by p ; more generally, knowledge of ${}_p\mathbf{Pic}_{S/k}$ is equivalent to that of $H^1(S, G)$ for every finite commutative algebraic group G annihilated by p^n , by virtue of the following theorem, which encompasses, in the case under consideration, both Kummer theory and that of Artin-Schreier:

Let G be a finite algebraic group over k , and $D(G) = \mathbf{Hom}_{k\text{-groups}}(G, \mathbf{G}_m)$ its *Cartier dual* (whose affine algebra is carried by the vector space dual to the affine algebra of G , i.e. by the hyperalgebra of G in the sense of Dieudonné-Cartier); then one has a canonical isomorphism:

$$(*) \quad H^1(S, G) \simeq \text{Hom}_{k\text{-groups}}(D(G), \mathbf{Pic}_{S/k}).$$

(N.B. S is a proper scheme over k algebraically closed, such that $H^0(S, \mathcal{O}_S) = k$). This formula can also be expressed by saying that the “true fundamental group” of S alluded to in no. 2, made abelian, is isomorphic to the projective limit of the $D(P_i)$, where P_i runs through the *finite* algebraic subgroups of $\mathbf{Pic}_{S/k}$, which will be denoted $T^\bullet(\mathbf{Pic}_{S/k})$. When S is an abelian variety, we have seen in 2.1 that this group is also isomorphic to the “true” Tate module $T_\bullet(S) = \varprojlim_n S$, and the isomorphism $(*)$ can then be written more strikingly as

$$\text{Ext}^1(A, G) \simeq \text{Hom}(D(G), B),$$

A being an abelian variety, B its dual, G a finite algebraic group over k . The results just indicated can moreover be generalized to the case where k is replaced by an arbitrary base prescheme, and to coefficient groups G other than finite groups.

7. Bibliography

- [1] A. GROTHENDIECK, Sur quelques points d’algèbre homologique, Tôhoku Math. J. **9** (1957) p. 119–221.
- [2] A. GROTHENDIECK, A general theory of fibre spaces with structure sheaf, University of Kansas, 1955.
- [3] A. GROTHENDIECK, Torsion homologique et sections rationnelles, Séminaire Chevalley, June 16, 1958.
- [4] S. LANG, Abelian varieties, Interscience tracts in pure and applied mathematics, no. 7, New York.
- [5] J-P. SERRE, Groupes algébriques et corps de classes, Actualités Scient. et Ind. no. 1264, Hermann, Paris, 1959.
- [6] J-P. SERRE, Groupes proalgébriques, Publications Mathématiques de l’IHES **7** (1960), p. 1–67.
- [7] J-P. SERRE, Espaces fibrés algébriques, Séminaire Chevalley, April 21, 1958.
- [8] J-P. SERRE, Quelques propriétés des variétés abéliennes en caractéristique p , Amer. J. Math. **80** (1958), p. 715–739.
- [9] J-P. SERRE, Sur la topologie des variétés algébriques en caractéristique p , Symposium Internacional de Topologia Algebraica, 1958.
- [10] J-P. SERRE, On the fundamental group of a unirational variety, J. London Math. Soc. **34** (1959), p. 481–484.